DOMAIN: Online Retail E-Commerce Fraud Detection System

BUSINESS PROBLEM

Online retail platforms like Amazon or Flipkart process millions of daily transactions. Some customers perform fraudulent activities such as fake returns, payment fraud, account takeovers, or using stolen credit cards. This causes financial loss, reputational damage, and operational inefficiencies.

Business Objective

- Predict fraudulent transactions in advance
- Reduce financial losses due to fraud
- Improve transaction security
- Protect genuine customers
- Take preventive actions such as:
  - Blocking suspicious transactions
  - Enabling additional verification (OTP / KYC)
  - Restricting high-risk accounts
  - Sending fraud alerts to customers

Business Success Criteria

- Reduce fraud rate (e.g., by 20–25%)
- Reduce chargeback losses
- Improve genuine transaction approval rate
- Increase customer trust
- Increase overall profitability

Assess Situation

Inventory of Resources

- Historical transaction data
- Customer profile data (location, purchase history)
- Payment method data (Credit card / UPI / COD)
- Device and IP address data
- Order and return history
- Existing IT infrastructure (cloud servers, databases)
- Cybersecurity and analytics team

Requirements

- Accurate fraud detection
- Real-time fraud prediction during checkout
- Fast system response
- Easy integration with payment gateway

Constraints

- Data privacy regulations (GDPR, PCI-DSS)
- Highly imbalanced dataset (very few fraud cases)
- Evolving fraud techniques
- Real-time latency limitations

Costs:
- Data storage and processing
- Model development and testing
- Cloud infrastructure
- Skilled cybersecurity professionals

Benefits:
- Reduced fraud losses
- Increased secure transactions
- Improved customer trust
- Higher long-term revenue

Determine Data Science Goals

Data Science Objective (Technical View)

Build a classification model to predict whether a transaction is:

- Fraudulent (Yes)
- Genuine (No)

Data Science Tasks

- Data preprocessing
- Feature engineering (transaction frequency, unusual spending patterns)
- Handle missing values
- Handle class imbalance (SMOTE / Class weighting)
- Train models

Produce Project Plan

Project Plan Overview

Stage	Activity	Duration
1	Business Understanding	1 week
2	Data Collection & Understanding	2 weeks
3	Data Cleaning & Preparation	2 weeks
4	Model Building	3 weeks
5	Model Evaluation	1 week
6	Deployment	1 week
7	Monitoring & Maintenance	Continuous

Resources Needed

- Data Scientists
- Software Engineers
- Cybersecurity Experts
- Domain Experts
- Cloud Infrastructure

Tools & Techniques

- Python (Pandas, NumPy, Scikit-learn)
- SQL
- Power BI / Tableau
- Machine Learning Algorithms
- Cloud Platforms (AWS / Azure / GCP)

Final Outcome

- Deployed Fraud Detection System
- Reduced financial losses
- Improved transaction security
- Enhanced customer trust
- Increased operational efficiency